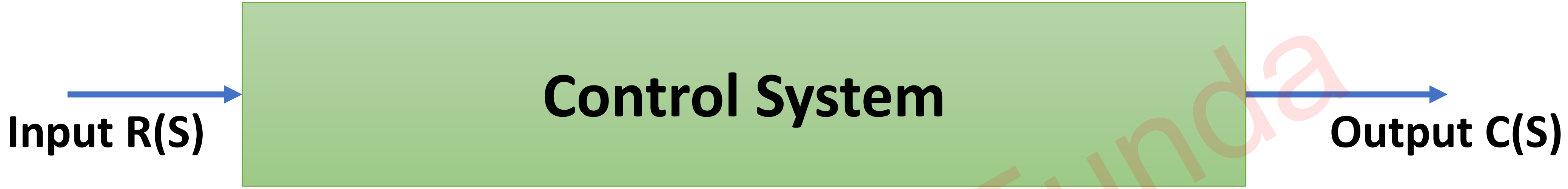
Transfer Function

Outlines

- Basics of Transfer Function
- Types of Transfer Function
- Properties of Transfer Function
- Advantages of Transfer Function
- Disadvantages of Transfer Function

Basics of Transfer Function

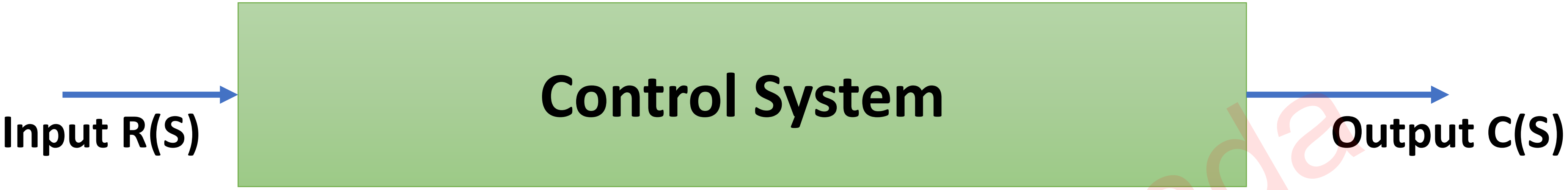
- Transfer Function gives the relationship between the input and the output of the system.



$$\Rightarrow \mathbf{T(s)} = \frac{\mathbf{C(S)}}{\mathbf{R(S)}} = \frac{\mathbf{N(S)}}{\mathbf{D(S)}}$$

- Here, The Roots of N(S) gives details of Zeros of System and The Roots of D(S) gives details of Poles of System.
- Transfer Function usually represented in form of frequency domain {Laplace Transforms (Continuous Time Systems), Z Transforms (Discrete Time Systems)}.

Types of Transfer Function

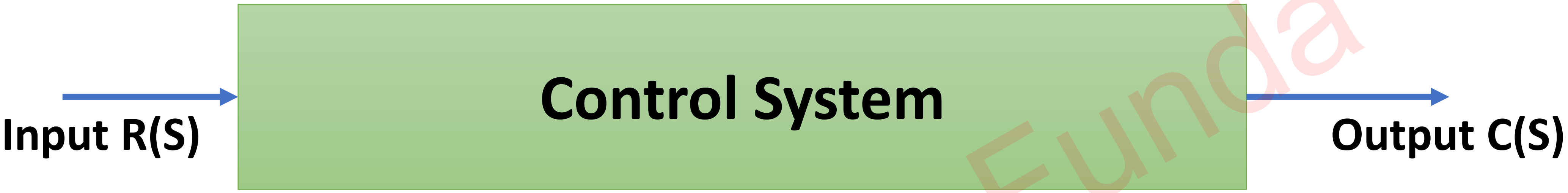


$$\Rightarrow \mathbf{T(s)} = \frac{\mathbf{C(S)}}{\mathbf{R(S)}} = \frac{\mathbf{N(S)}}{\mathbf{D(S)}}$$

- **Proper Transfer Function**
  - **Poles P > Zeros Z {Order of D(S) is Greater then Order of N(S).}**
- **Improper Transfer Function**
  - **Poles P < Zeros Z {Order of D(S) is less then Order of N(S).}**

Properties of Transfer Function

- The Transfer Function of a system is the Laplace Transform of its Impulse Response for Zero Initial Conditions.



$$\Rightarrow \mathbf{T(s)} = \frac{\mathbf{C(S)}}{\mathbf{R(S)}} = \frac{\mathbf{N(S)}}{\mathbf{D(S)}}$$

- The Transfer Function can be determined from Input – Output pair by taking ratio of Laplace Output to Laplace of Input.
- Practically, Transfer Function is Independent of the Inputs to the System.
- The Transfer Function can be defined only for LTI (Linear Time Invariant) System. For Non Linear System, response of the system changes with respect to time.
- Poles and Zeros of the System can be identified by Transfer Function.
- Characteristics and Stability of the System can be identified by Transfer Function.

Advantages of Transfer Function

- Transfer Function is a mathematical model that gives the gain of the given system.
- Integral and Differential Equations are converted to simple algebraic equation.
- From Transfer Function, Output of system can be identified for any Inputs.
- Transfer Function is dependent on system only, not on Inputs.
- Using Transfer Function, Poles, Zeros, Stability and Characteristics can be Identified.

Disadvantages of Transfer Function

- Transfer Function is only valid for LTI Systems.
- Transfer Function doesn’t account for Initial Conditions.
- Transfer Function doesn’t give any idea about how the present output is progressing.

Transfer Function From Differential Equation

Question 1 – If the Differential Equation is given by:

$$\Rightarrow \frac{d^2y}{dt^2} + \frac{dy}{dt} + y = 6x \qquad \Rightarrow T(s) = \frac{Y(s)}{X(s)}$$

Where x is Input and y is Output.

Find, The Transfer Function of the given Differential Equation.

~> Apply Laplace x'form to differential eq.^{n}

$$\Rightarrow [s^2 Y(s) - sY(0) - Y'(0)] + [sY(s) - Y(0)] + Y(s) = 6X(s)$$

~> Apply zero initial cond.^{n}

$$\Rightarrow Y(s)[s^2 + s + 1] = 6X(s)$$

$$\Rightarrow T(s) = \frac{Y(s)}{X(s)} = \frac{6}{s^2 + s + 1}$$

Question 2 –Find The Transfer Function of the given Differential Equation.

$$\Rightarrow Y'' = 5, Y(0) = 1, Y'(0) = 2 \quad — 1$$

$$\Rightarrow X' = e^{-t}, X(0) = 0 \quad — 2$$

Where Y is Output and X is Input.

→ Apply Laplace x'form to eqⁿ 1

$$\Rightarrow [s^2 y(s) - s y(0) - y'(0)] = 5/s$$

$$\Rightarrow [s^2 y(s) - s(1) - 2] = 5/s$$

$$\Rightarrow s^2 y(s) = \frac{5}{s} + s + 2$$

$$\Rightarrow s^2 y(s) = \frac{s^2 + 2s + 5}{s}$$

$$\Rightarrow y(s) = \frac{s^2 + 2s + 5}{s^3} \quad — A$$

→ Apply Laplace x'form to eqⁿ 2

$$\Rightarrow [s x(s) - x(0)] = \frac{1}{s+1}$$

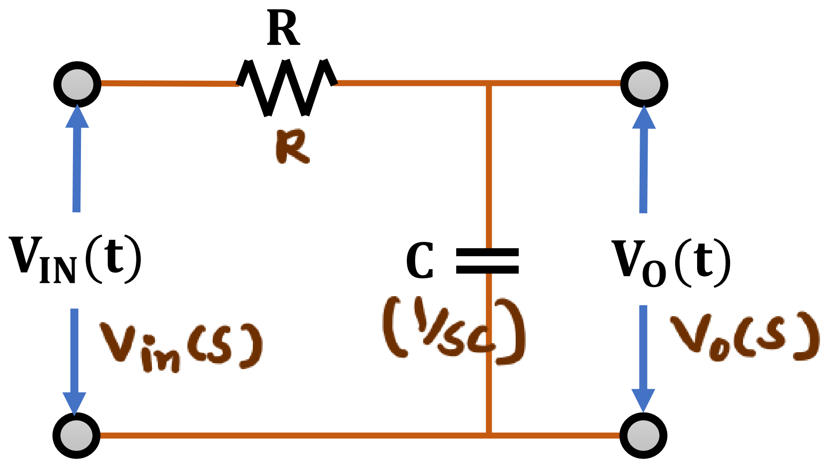
$$\Rightarrow x(s) = \frac{1}{s(s+1)} \quad — B$$

$$\rightarrow T(s) = \frac{y(s)}{x(s)} = \frac{s^2 + 2s + 5/s^{\cancel{3}\,2}}{1/\cancel{s}(s+1)}$$

$$= \frac{(s+1)(s^2 + 2s + 5)}{s^2}$$

Transfer Function of RC Circuits

Question 1 – Find the Transfer Function of a given (LPF) RC Circuit.



→ Zero initial cond.ⁿ

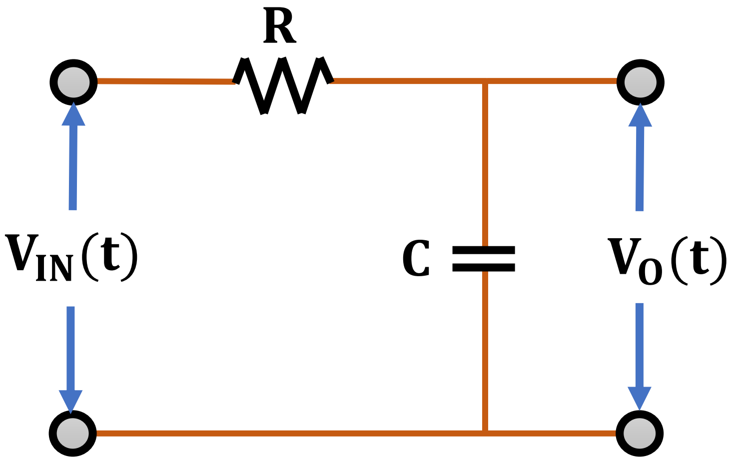
$t=0, \; V_c = 0$.

→ Transfer function $T(s) = \dfrac{V_o(s)}{V_{in}(s)}$

→ Apply voltage divider rule.

$$\Rightarrow V_o(s) = V_{in}(s)\left(\frac{1/sC}{R + 1/sC}\right)$$

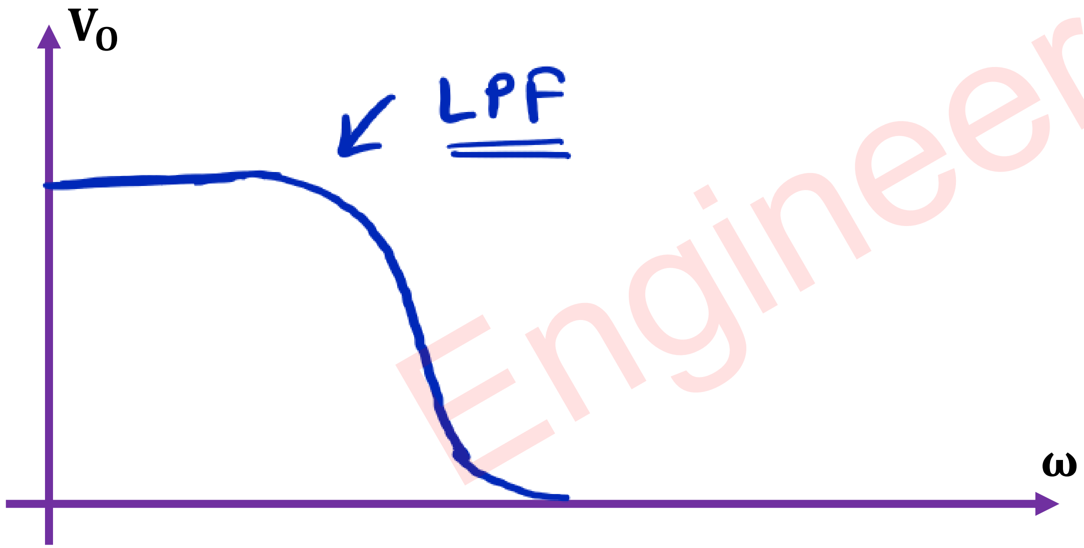
$$\Rightarrow \frac{V_o(s)}{V_{in}(s)} = T(s) = \frac{1}{1+RsC}$$



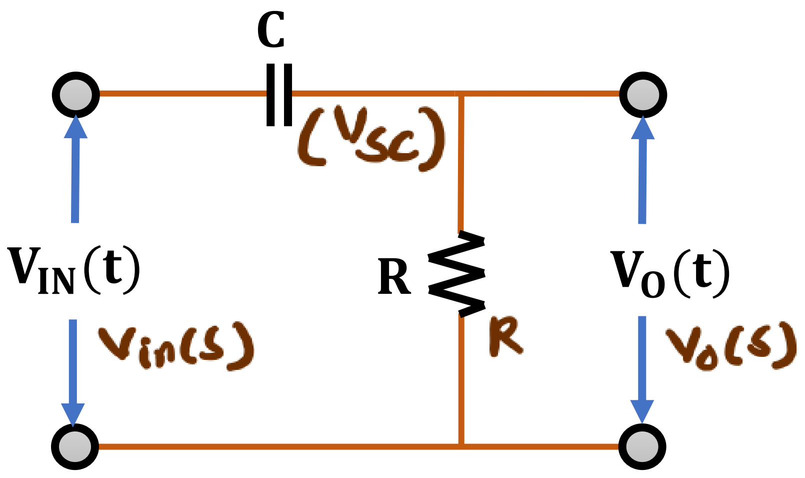
$\rightarrow X_c = 1/sC = \dfrac{1}{j\omega C}$

$\rightarrow$ at $\omega \rightarrow$ low, X_c = high (O.C.), I/P $\rightarrow$ O/P

at $\omega \rightarrow$ high, X_c = low (S.C.), O/P $\rightarrow$ 0



Question 2 – Find the Transfer Function of a given (HPF) RC Circuit.



~ Zero initial cond.^{n}

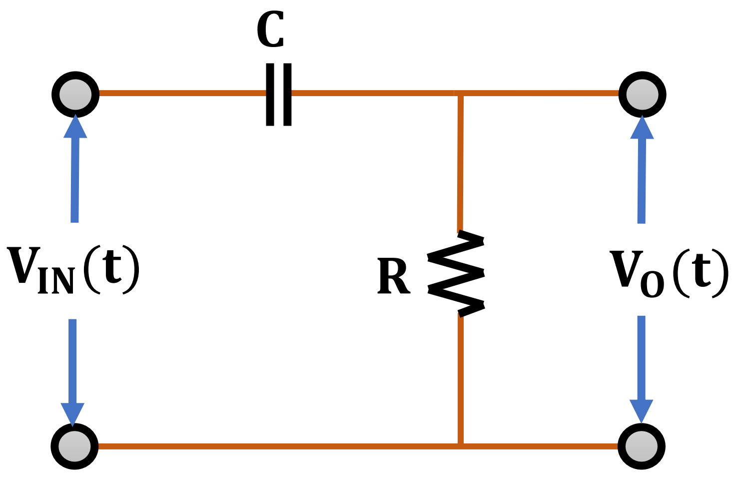
$t = 0, \; V_c = 0$

~ Transfer function $T(s) = \dfrac{V_o(s)}{V_{in}(s)}$

~ Apply voltage divider rule.

$$\Rightarrow V_o(s) = V_{in}(s)\left(\frac{R}{R + 1/sC}\right)$$

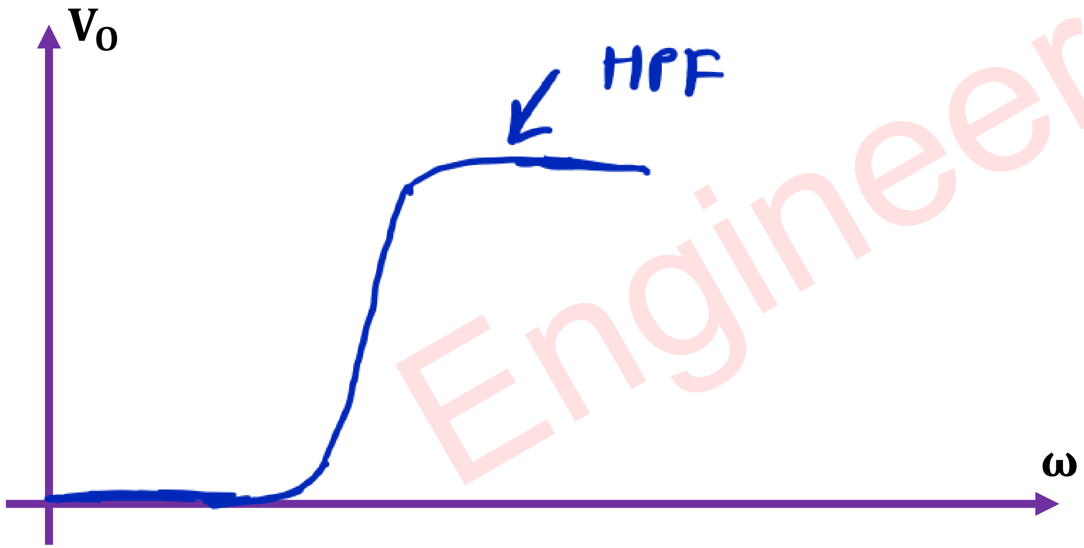
$$\Rightarrow \frac{V_o(s)}{V_{in}(s)} = T(s) = \frac{RsC}{1 + RsC}$$



→ $X_C = \frac{1}{sC} = \frac{1}{j\omega C}$

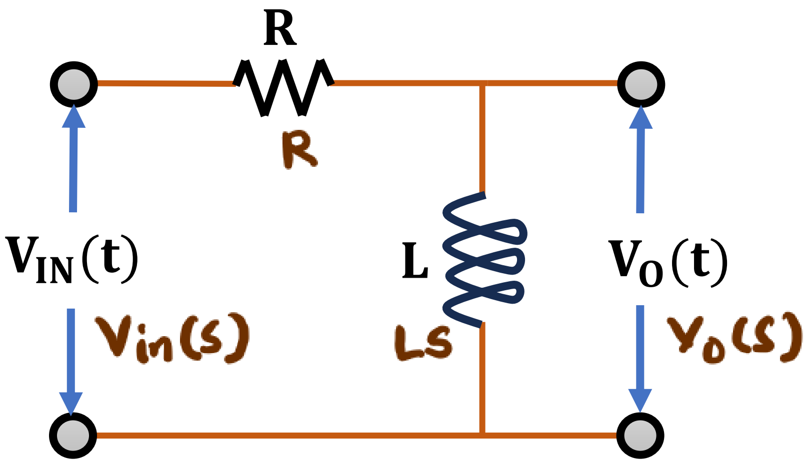
→ $\omega \to$ low, $X_C =$ high (O.C.), $V_o \to 0$

$\omega \to$ high, $X_C =$ low (S.C.), $V_o \to V_{in}$



Transfer Function of RL Circuits

Question 1 – Find the Transfer Function of a given (HPF) RL Circuit.



~> Zero initial cond.^{n}

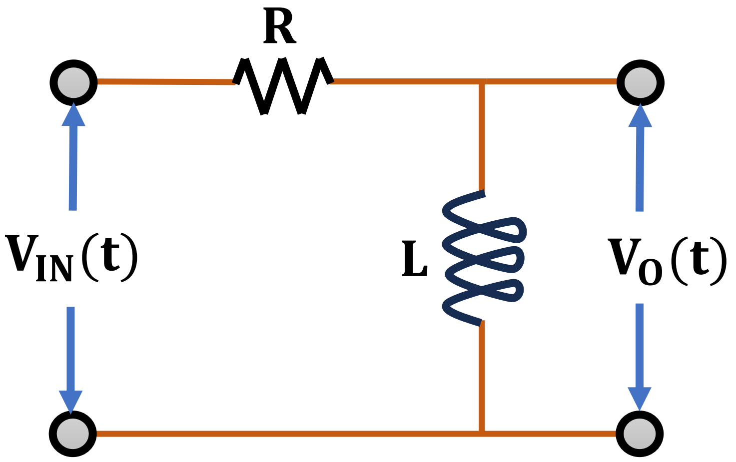
$t=0, \ I_L = 0$

~> Transfer function $T(s) = \dfrac{V_o(s)}{V_{in}(s)}$

~> Apply voltage divider rule.

$$\Rightarrow V_o(s) = V_{in}(s)\left(\frac{LS}{R+LS}\right)$$

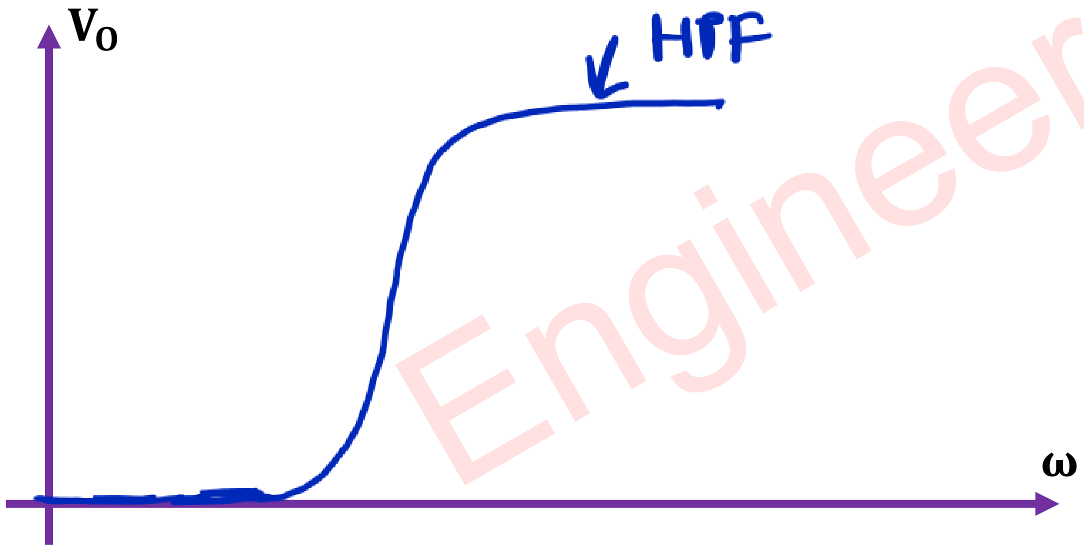
$$\Rightarrow \frac{V_o(s)}{V_{in}(s)} = T(s) = \frac{LS}{R+LS}$$



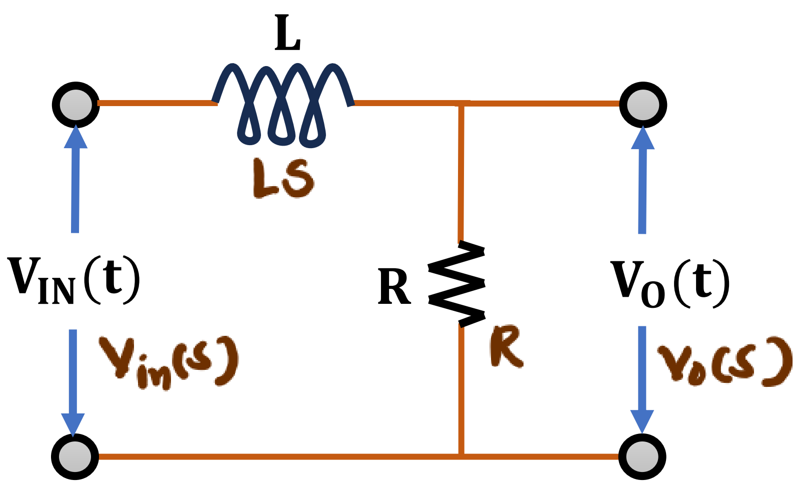
$\rightsquigarrow X_L = sL = j\omega L$

$\rightsquigarrow \omega \to \text{low}, \; X_L = \text{low (s.c.)}, \; V_o \to 0$

$\omega \to \text{high}, \; X_L = \text{high (o.c.)}, \; V_{in} \to V_o.$



Question 2 – Find the Transfer Function of a given (LPF) RL Circuit.



~ Zero initial cond.^{n}

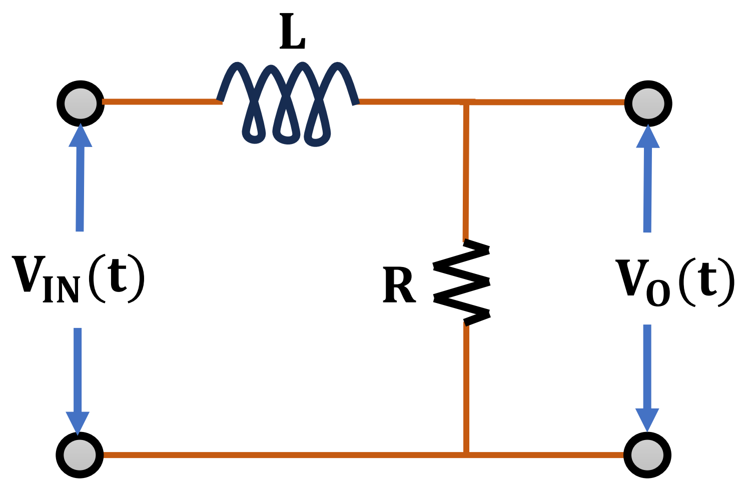
$t=0,\ I_L=0$

~ Transfer function $T(S) = \dfrac{V_o(S)}{V_{in}(S)}$

~ Apply Voltage divider rule.

$$\Rightarrow V_o(S) = V_{in}(S)\left(\frac{R}{R+LS}\right)$$

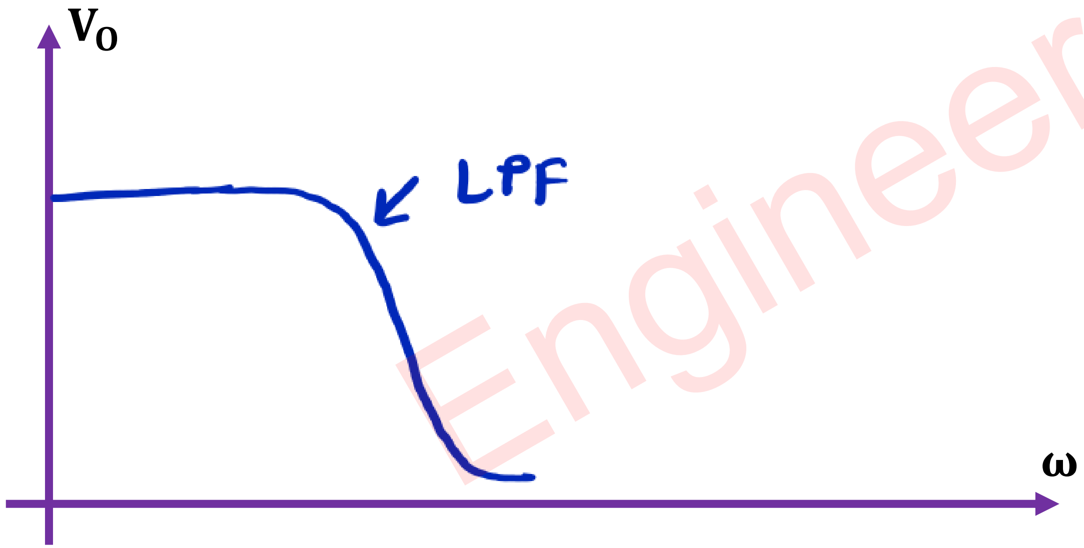
$$\Rightarrow \frac{V_o(S)}{V_{in}(S)} = T(S) = \frac{R}{R+LS}$$



$\rightsquigarrow X_L = sL = j\omega L$

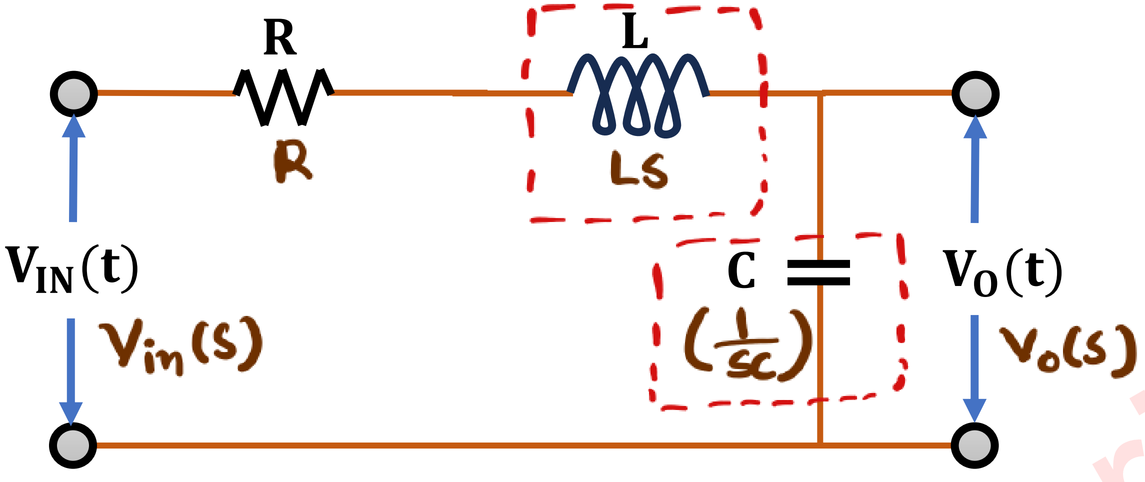
$\rightsquigarrow \omega \to \text{low}$, $X_L = \text{low (s.c)}$, $V_{in} \to V_o$

$\omega \to \text{high}$, $X_L = \text{high (o.c.)}$, $V_o \to 0$



Transfer Function of RLC Circuits

Question 1 – Find the Transfer Function of a given RLC Circuit.



⤳ Apply zero initial cond.^{n}

$t=0,\ V_c = 0,\ I_L = 0$

⤳ Transfer function

$$T(S) = \frac{V_o(S)}{V_{in}(S)}$$

⤳ Apply voltage divider rule.

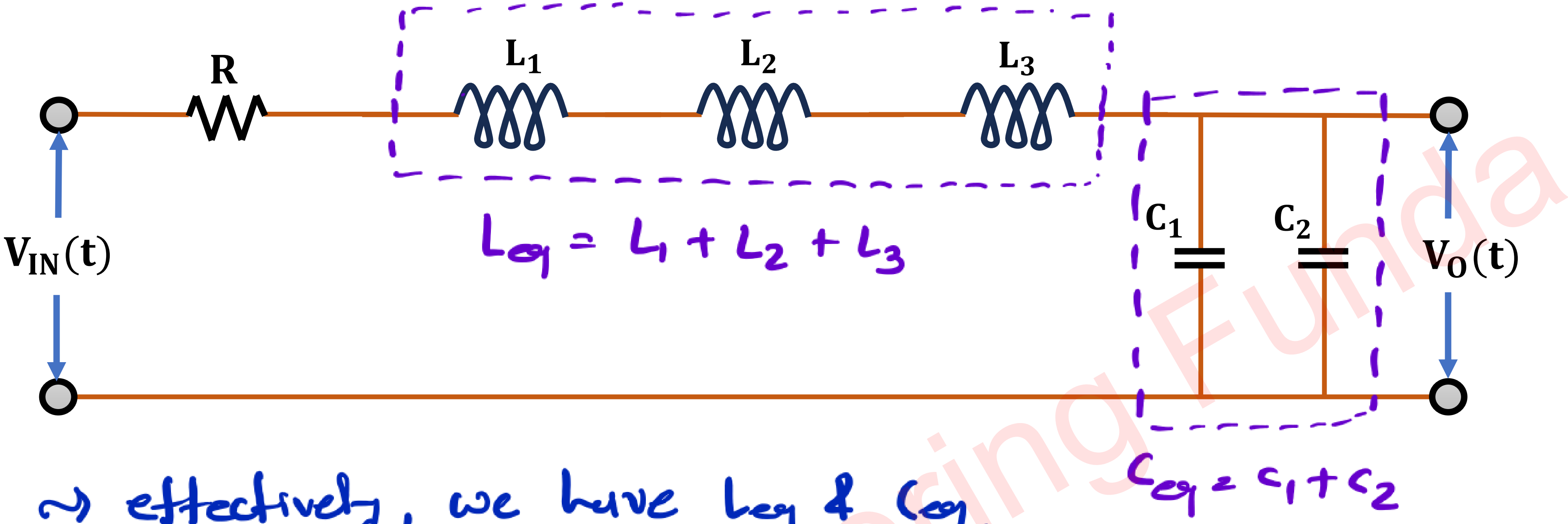
$$\Rightarrow V_o(S) = V_{in}(S)\left[\frac{1/SC}{R + LS + 1/SC}\right]$$

$$\Rightarrow \frac{V_o(S)}{V_{in}(S)} = T(S) = \frac{1}{LC\,S^2 + RC\,S + 1}$$

S^2 means Order = 2

⤳ In circuit, we have two effective elements which can stores energy. So order of system = 2

Question 2 – Find the Order of the Circuit.



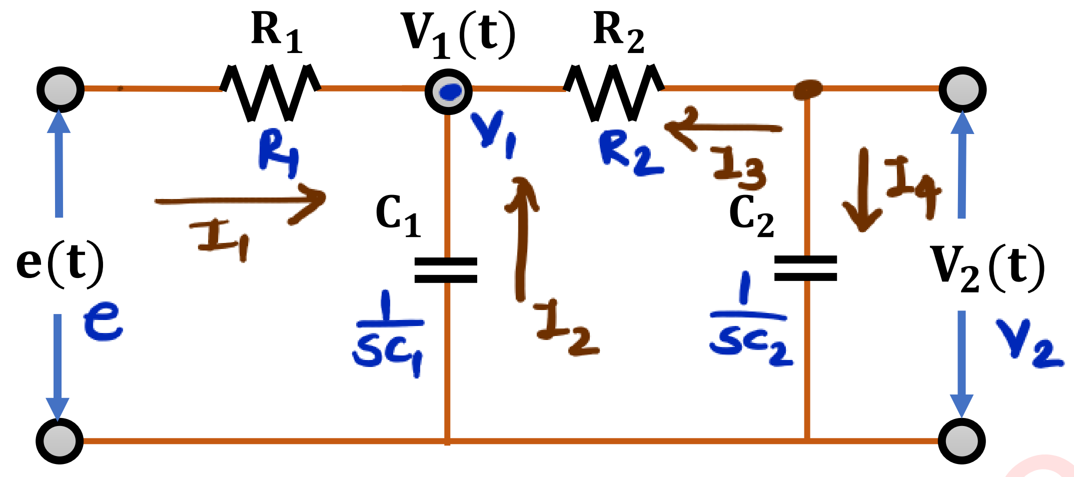
→ effectively, we have L_{eq} & C_{eq}.
So, order of the system is 2.

→ $T(s) = \dfrac{V_o(s)}{V_{in}(s)} = \dfrac{1}{L_{eq}C_{eq}s^2 + RC_{eq}s + 1}$

s^2 term

Transfer Function of Electrical N/W

Question – Find the Transfer Function of the Electrical Network. $\frac{V_2}{e} = ?$



~ Apply KCL at Node V_1

$$\Rightarrow I_1 + I_2 + I_3 = 0$$

$$\Rightarrow \frac{V_1 - e}{R_1} + \frac{V_1}{(1/sC_1)} + \frac{V_1 - V_2}{R_2} = 0$$

$$\Rightarrow V_1\left[\frac{1}{R_1} + \frac{1}{R_2} + sC_1\right] = \frac{e}{R_1} + \frac{V_2}{R_2}$$

$$\Rightarrow V_1\left[R_1 + R_2 + sC_1R_1R_2\right] = eR_2 + V_2R_1$$

$$\Rightarrow V_1 = \frac{eR_2 + V_2R_1}{R_1 + R_2 + sC_1R_1R_2} \quad \text{—— (1)}$$

~ Apply KCL at Node V_2

$$\Rightarrow I_3 + I_4 = 0$$

$$\Rightarrow \frac{V_1 - V_2}{R_2} + \frac{-V_2}{(1/sC_2)} = 0$$

$$\Rightarrow \frac{V_1}{R_2} = V_2\left[\frac{1}{R_2} + sC_2\right]$$

$$\Rightarrow V_1 = V_2\left[1 + sC_2R_2\right] \quad \text{— } (2)$$

$\rightsquigarrow$ Put (1) into (2)

$$\Rightarrow \frac{eR_2 + V_2R_1}{R_1 + R_2 + sC_1R_1R_2} \times \frac{1}{1+sC_2R_2} = V_2$$

$$\Rightarrow \frac{eR_2}{(1+sC_2R_2)(R_1+R_2+sC_1R_1R_2)} = V_2\left[1 - \frac{R_1}{(1+sC_2R_2)(R_1+R_2+sC_1R_1R_2)}\right]$$

$$\Rightarrow eR_2 = V_2\left[(1+sC_2R_2)(R_1+R_2+sC_1R_1R_2) - R_1\right]$$

$$\Rightarrow \boxed{\frac{V_2}{e} = \frac{R_2}{(1+sC_2R_2)(R_1+R_2+sC_1R_1R_2) - R_2}}$$